

SYNCHRONOUS MACHINE

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$$f = \frac{PN}{120} \text{ Hz}$$

$$50 = \frac{PN}{120}$$

$$PN = 6000$$

armature wdg → stator (or) Rotor means both. because of generation of ac
armature wdg → Rotor (dc m/c) (DAR)

Field (or) cond^r both can rotate.

247 MVA, 0.85 PF lag, 3-φ Gen^r, 50 Hz, 3000 rpm

armature wdg:- 3-φ AC, 15.75 kV, 9050 A, double star. → on stator.
(Parallel line)

Insulation, High current, cooling are difficult in above.

Field wdg:- 2600 A dc, 310 V dc. → on rotor.

5 Hz → Gives lamps flicker effect.

100 Hz → Gives xⁿ problem.

50 Hz → Reliable. (std.)

(freq. ↑ size ↓)

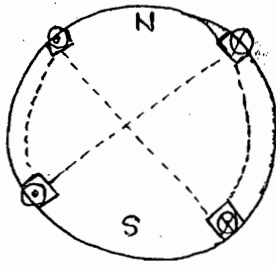
* cylindrical rotor / Round Rotor / Non-salient

* turbogenerator - Ni cd (3000 rpm) No lamination
Rotor

Relative velocity 0.

- 115 cm diameter (periphery) ↓ area for poles ↓.

Cylindrical Rotor/Non-salient

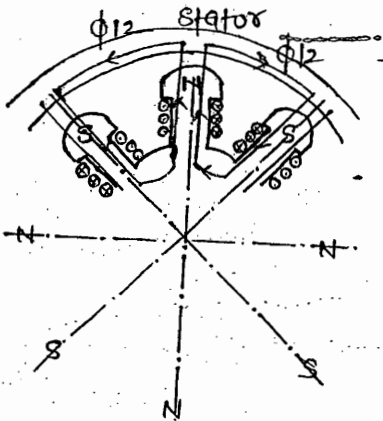


Poles: $\frac{1}{3}$ rd

field wdg = $\frac{2}{3}$ rd

$P \neq 4$

Salient pole rotor (salient - projected)



* leakage flux doesn't take place in the power transfer. like electric choke.

* mutual flux is responsible for power transfer.

Sinusoidal waveform → This only producing the ~~REF~~ RMF

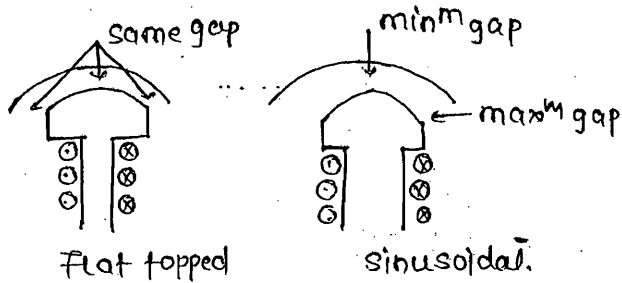
* This can be expressed in complex (phasors).

BLV
 ↳ Not sinusoidal.
 ↳ must be sinusoidal.

flux per pole - Total no. of ~~flux~~ pole flux line that an observer were find in total N-space. (1-pole pitch)

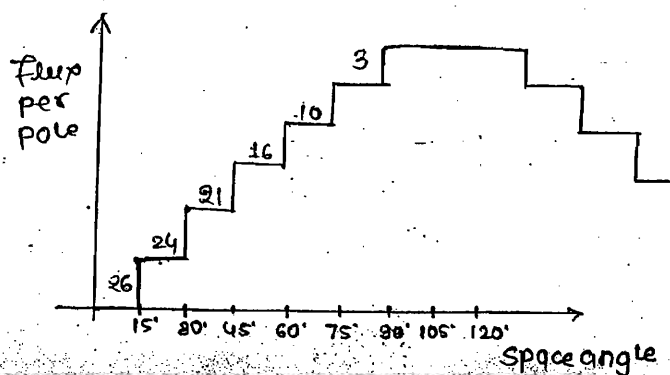
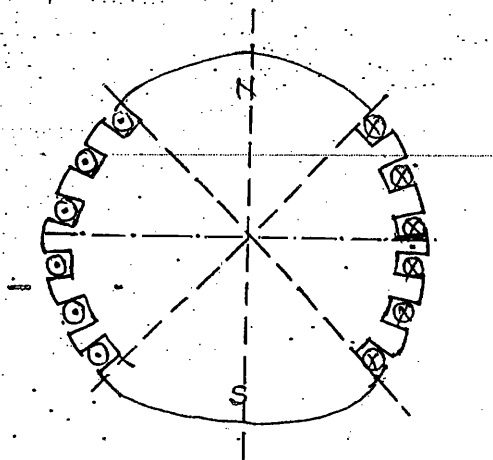
Sinusoidal flux density distribution in the air gap:-

- (1) The req. waveform of generated vol. is sinusoidal.
- (2) A sinusoidal vol. may only be generated if the radial flux density in the air gap is sinusoidal.



- (3) In a salient pole rotor the field wdg is concentric but concentrated & therefore the field mmf in the air gap of the pole arc is constant. Hence a sinusoidal flux density distribution may be obtained by shaping the pole such that the air gap at the center of pole kept min & goes on increasing as one moves away from the center of the pole.

Of course the variation in the air gap should be compatible with the desired sinusoid. (Ref above fig)



In a cylindrical rotor the field wdg is concentric but distributed in space in no. of slots. The air gap however is uniform & therefore the reluctance of airgap is constant.

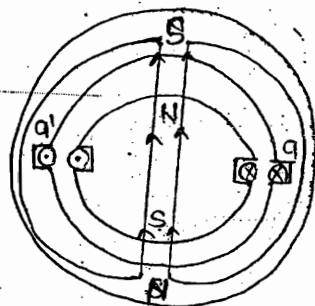
Therefore a sinusoidal flux density may only be obtained with a sinusoidal field mmf wave.

This may be obtained by providing max^m no. of turns in the slots adjacent to the interpolar axis & the no. of turns in the subsequent slots goes on decreasing as one approaches the pole. Such that the min^m no. of turns is provided in the slots adjacent to the pole.

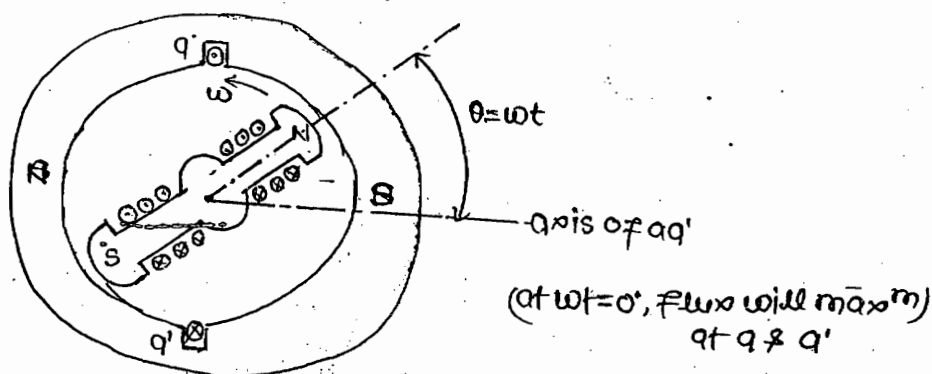
Ofcourse the variation in the no. of turns the diff slots should be compatible with the desired flux density distribution.

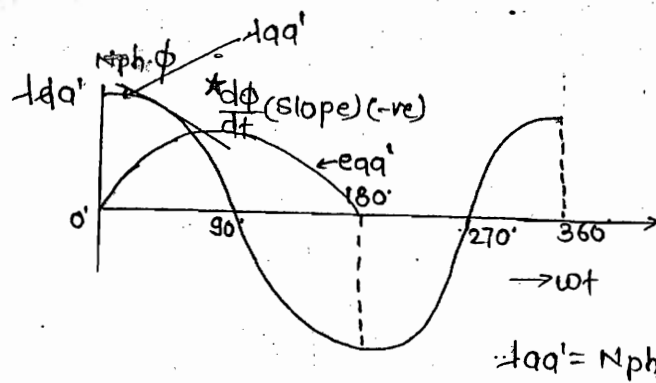
Such a provision results into a stepped field mmf wave that approximate a sinusoid.

A perfect sinusoidal mmf may only be obtained if the no. of field slots is infinite.



Induced EMF →





P = No. of pole, N = Speed in rpm.

$$f = \frac{PN}{1000} \text{ Hz}$$

ϕ = flux/pole; = $B_{avg} \times \text{Area/pole}$

$$= \left(\frac{2}{\pi} B_m\right) \times \frac{\pi D L}{P} \text{ wb/pole}$$

N_{ph} = No. of turns/phase.

$$e_{aa'} = \pm \frac{d}{dt} (\lambda_{aa'})$$

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$$\begin{aligned} e_{aa'} &= \frac{d}{dt} (N_{ph} \phi \cos \omega t) = - \left[N_{ph} \phi \frac{d}{dt} (\cos \omega t) + N_{ph} \cos \omega t \frac{d\phi}{dt} \right] \\ &= - N_{ph} \phi (-\sin \omega t) \omega t = - \left[N_{ph} \phi \omega (-\sin \omega t) + N_{ph} \cos \omega t \frac{d\phi}{dt} \right] \\ &= N_{ph} \phi \omega \sin \omega t \\ &= N_{ph} \phi \omega \sin \omega t = \left[N_{ph} \phi \omega \sin \omega t - N_{ph} \cos \omega t \frac{d\phi}{dt} \right] \end{aligned}$$

$e_{aa'}$ = speed voltage

$e_{aa'} = \underbrace{N_{ph} \phi \omega \sin \omega t}_{\text{Speed voltage}} - \underbrace{N_{ph} \cos \omega t \frac{d\phi}{dt}}_{\text{Xmex voltage}}$

Since all synchronous m/c have there field excited by dc, $\frac{d\phi}{dt} = 0$

$$e_{aa'} = N_{ph} \phi \omega \sin \omega t = \underbrace{N_{ph} \phi \omega}_{\text{max}^m \text{ value}} \cos(\omega t - 90^\circ)$$

Therefore RMS $E_{ph} = \frac{N_{ph} \cdot \phi \omega}{\sqrt{2}}$

$$= \frac{N_{ph} \cdot \phi \cdot (2\pi f)}{\sqrt{2}}$$

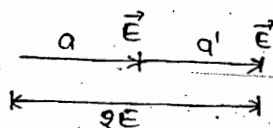
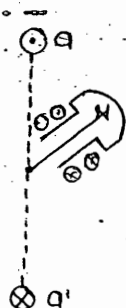
$$E_{ph} = \sqrt{2} \pi f \phi N_{ph}$$

$$E_{ph} = \sqrt{2} \pi f \phi \cdot N_{ph} \text{ volts/ph.}$$

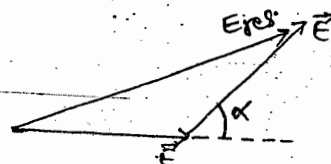
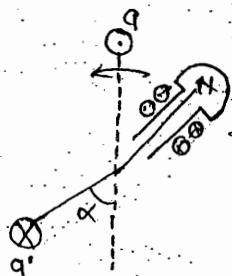
Valid for full pitched concentrated winding

DATE-12/11/14

Chorded winding →



Full pitched wdg



$$E_{res} = 2E \cos \frac{\alpha}{2}$$

CHORDED wdg.

chording factor/coil span factor/pitch factor

$$k_c = \cos \frac{\alpha}{2} \text{ for fundamental}$$

for nth harmonic voltage
replace $(\alpha = n\alpha)$

$$\text{Then } k_c(n) = \cos \frac{n\alpha}{2}$$

To eliminate nth harmonic

$$\frac{n\alpha}{2} = 90^\circ$$

$$\alpha = \frac{180^\circ}{n}$$

However, practical value of α is 30° elect. ($\alpha = 30^\circ$)

If m -phase sys. the $2m \pm 1$ (Belt harmonics)

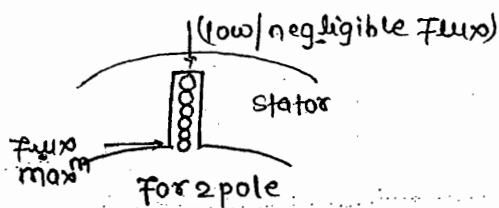
$m=3$ (3- ϕ sys) 5th & 7th harmonics are belt harmonics

* chording reduces the harmonics

* If $\alpha = 60^\circ$, more cu will be needed, so cost increases here.

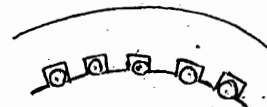
$$E_{ph} = K_c \sqrt{2} \pi f \phi N_{ph} \text{ Vol/\phi}$$

(Valid for chorded but concentrated wdg.)



* Iron increases because of depth increasing.

* Saw Needed more iron

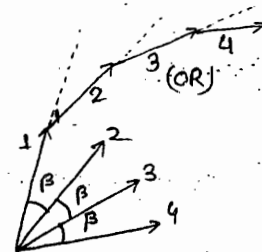
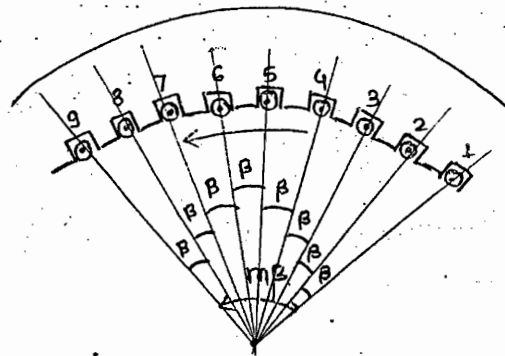


* Iron reduces & Flux is same for all the cond.

(Good design)

* Iron needed reduces.

* Distributed winding $\rightarrow$



β = angle between adjacent slots.

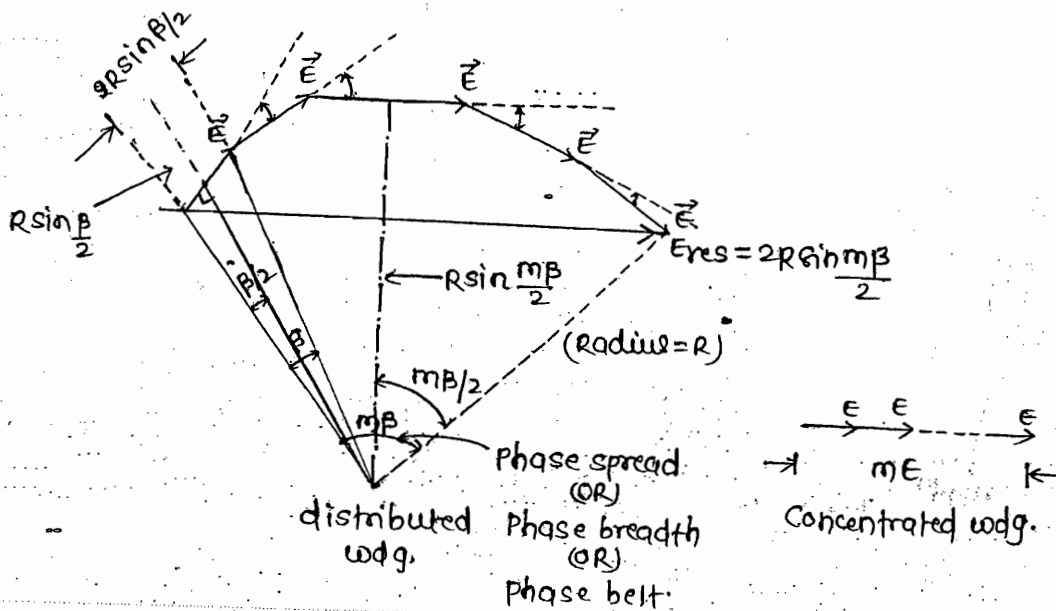
$$= \frac{360^\circ}{\text{No. of slots}} \text{ mech.}$$

$$= \frac{P}{2} \times \frac{360^\circ}{\text{No. of slots}} \text{ mech. ele.}$$

$$= \frac{180^\circ}{\text{No. of slots per pole}} \text{ elec.}$$

$$m = (\text{No. of slots per pole}) \text{ per phase} = \text{No. of slots per pole phase}$$

$$= (\text{no. of slots/pole}) / \text{phase}$$



$$\text{distribution factor, } k_d = \frac{2R \sin \frac{\alpha}{2}}{m_F}$$

$$\frac{= 2R \sin m\beta}{m \times 2R \sin \frac{\beta}{2}}$$

$$kd = \frac{\sin \frac{m\beta}{2}}{\sin \beta/2}$$

~~For Fundamental..~~

* For n th harmonics;

Replace β by $n\beta$

$$k_d(n) = \frac{\sin \cdot m(n\beta)}{2} \cdot \frac{1}{m \sin(\frac{n\beta}{2})}$$

*** distributed & chorded wdg is used mainly for reduction of harmonics.

approx k_d for fundamental vol.

since $\frac{\beta}{2}$ is very small

$$\sin \frac{\beta}{2} \approx \frac{\beta}{2}$$

$$\text{then } k_d = \frac{\sin \frac{m\beta}{2}}{\frac{m\beta}{2}}$$

$$\text{approx } k_d = \frac{\sin \frac{\text{Phase spread}}{2}}{\frac{\text{Phase spread}}{2}}$$

$$E_{ph} = k_c k_d \times \sqrt{2} \pi f \phi N_{ph} \text{ volts}/\phi$$

$$= k_w \sqrt{2} \pi f \phi N_{ph} \text{ V}/\phi$$

valid for chorded & distributed wdg.

where $k_w = k_c k_d = \text{winding factor}$.

Que → A 3φ 2p 3000 rpm λ connected cylindrical rotor turbogenerator has the following data

No. of slots = 60, max^m flux density in air gap = 1.32T

mean air gap dia = 1.12m, effective axial length = 3m,

No. of turns/φ = 10

Soln → Cal. the line to line vol. on NL if the coil span is 150°

$$\alpha = 180^\circ - 150^\circ$$

$$= 30^\circ \text{ ele.}$$

$$k_c = \cos \frac{30^\circ}{2} = 0.9659$$

$$\beta = \frac{180^\circ - 360^\circ}{60} \text{ mech.}$$

$$= \frac{p}{2} \times 6 \text{ ele.} = \frac{2}{2} \times 6 \text{ ele.}$$

$$\beta = 6^\circ \text{ ele.}$$

$$m = \frac{60/2}{3} = 10$$

$$k_d = \frac{\sin \frac{10 \times 6}{2}}{10 \sin \frac{6}{2}}$$

$$= 0.9554$$

$$\phi = \left(\frac{2}{\pi} \times 1.32 \right) \times \left(\frac{\pi \times 1.12 \times 3}{2} \right)$$

$$\phi = 4.4352 \text{ wb/pole}$$

$$E_{ph} = k_c k_d \times \sqrt{2} \pi f \phi N_{ph}$$

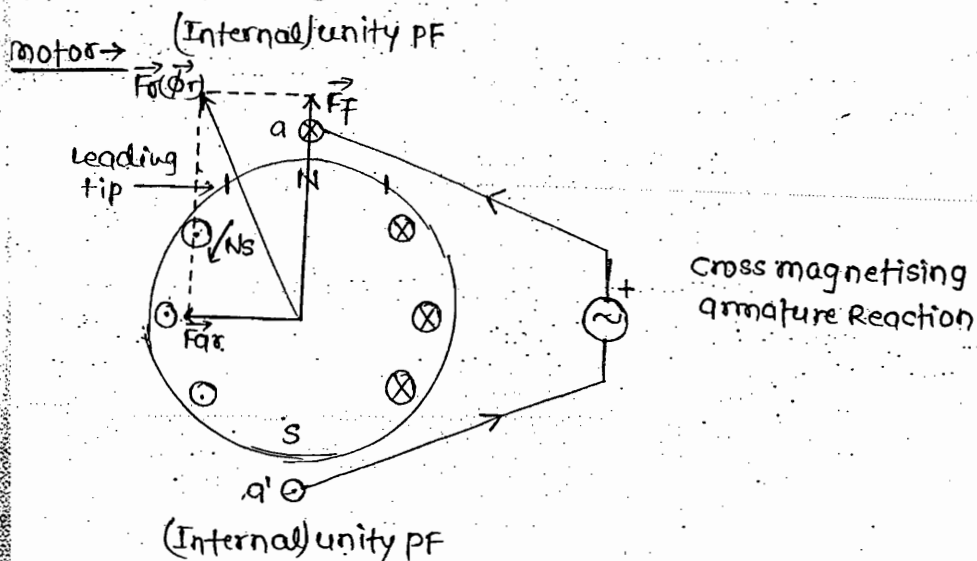
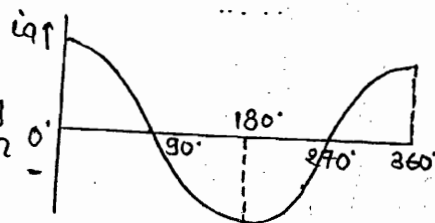
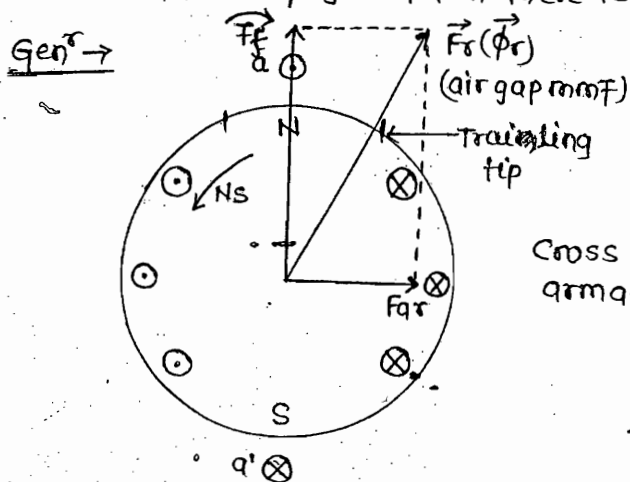
$$= 0.965 \times 0.9554 \times \sqrt{2} \times \pi \times 50 \times 4.4352 \times 10$$

$$= 9092.13 \text{ volts per star phase (line to neu.)}$$

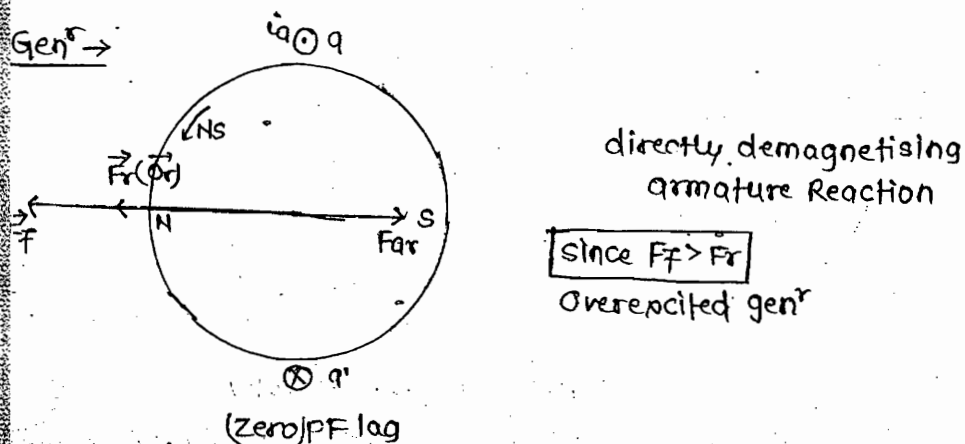
$$E_{(L-L)} = \sqrt{3} E_{ph} \approx 15.75 \text{ kV}$$

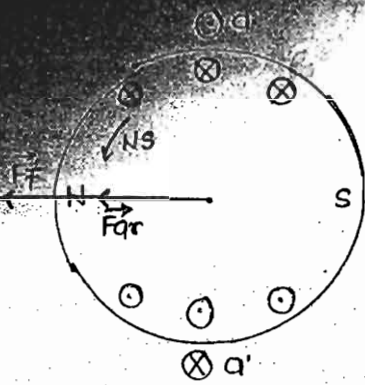
Armature Reaction →

- * The effect of arm. mmf on the main field is called arm. reaction.
- * Behind the every gen^r action there is motoring action.



- * Concentration of the flux is in the trailing tip at gen^r action & in the motor the concentration is in the leading tip.



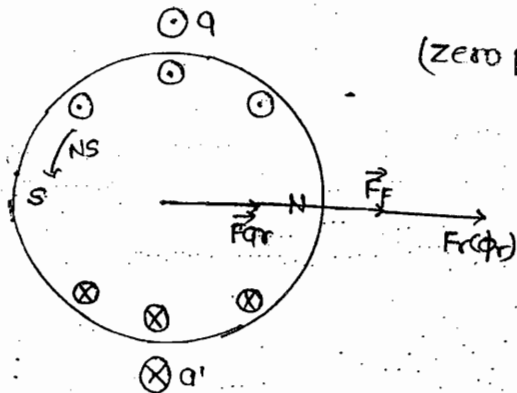


directly magnetising
armature reaction.

$$F_f < F_r$$

$\therefore$ Under excited motor

(zero) PF lag.

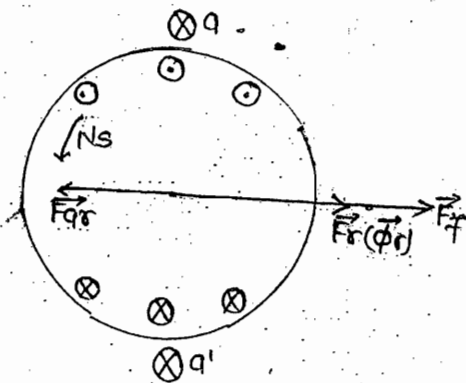


(zero PF lead)

directly magnetising
armature reaction

$$\text{Since } F_f < F_r$$

Underexcited gen^r



directly demagnetising
armature reaction

$$\text{Since } F_f > F_r$$

Overexcited motor